

Advanced Deep Learning Project Group 15

Adesijibomi Aderinto Oghenero Jeremiah Opia

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Objectives

General information about the project.

Project goals:

- ▶ Binary sentimental classification between hate and non hate speech.

Data Sources:

- ▶ Automated Hate Speech Detection and the Problem of Offensive Language (MrMorj 2022)
- ▶ DistilBert pretrained model (Sanh et al. 2019)
- ▶ Test evaluation data (Yer 2020)

Methodology

Implementation of a learning algorithm for binary classification capable of distinguishing clear and cloudy skies.

- ▶ Text Preprocessing: Removes;
 - ▶ Removes Emails
 - ▶ Removes IP addresses
 - ▶ Removes Punctuation
 - ▶ Removes Numbers
 - ▶ Removes stopwords using NLTK
 - ▶ Tokenizes words.

```
def preprocess_pandas(data, text_col='review'):
    data = data.copy()
    data[text_col] = data[text_col].str.lower()
    data[text_col] = data[text_col].replace(r'[a-zA-Z0-9._%+~]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}', '', regex=True)
    data[text_col] = data[text_col].replace(r'((25[0-5]|2[0-4][0-9]|01?[0-9]?[0-9]?)(\.|$)){4}', '', regex=True)
    data[text_col] = data[text_col].replace(r'^\w\s', '', regex=True)
    data[text_col] = data[text_col].replace(r'\d+', '', regex=True)

    stop_words = set(stopwords.words('english'))
    data[text_col] = data[text_col].apply(lambda x: " ".join(
        [word for word in word_tokenize(x) if word not in stop_words]
    ))
    return data
```

Figure 1: Data preprocessing

2. Model Training and Logging

- ▶ Custom Bidirectional LSTM model using Amazon hate speech data:
 - ▶ Uses Tokenizer and pad_sequences.
 - ▶ Train/val/test split.
 - ▶ Simple architecture: Embedding → BiLSTM → Dense.
 - ▶ Uses binary_crossentropy, tracks accuracy.
- ▶ Logs:
 - ▶ Loss/accuracy (train & val) to TensorBoard.
 - ▶ Confusion matrix
 - ▶ Word & sentence embeddings
 - ▶ Text predictions with probabilities

